

Library Escape: User-Centered Design of an Information Literacy Game

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ABSTRACT

The number of digital games for educational purposes has grown rapidly, and there is a potential for libraries to utilize them in information literacy (IL) education. Scholars have called for rigor in both the educational effectiveness and game-play experience in educational game design. One way to achieve this is to adopt a user-centered approach. Therefore, this article describes the user-centered design process of an IL game that actively involved potential users. A participatory design workshop and subsequent user evaluation yielded low- and high-fidelity prototypes. Insights on the complexities of educational game design were uncovered, including being accommodating yet pragmatic toward user input, keeping the balance between enjoyment and learning, and being guided yet flexible in using theoretical frameworks. Future directions for this IL game and general educational game design are also discussed.

The Internet has brought about the proliferation of and increased access to information. The ability to seek, locate, and navigate information effectively has thus become important. This is the very skill that information literacy (IL) provides: to allow people, whether students, knowledge workers, or the general public, to separate the wheat from the chaff in the quest to obtain high-quality content (Julien and Hoffman 2008). Librarians have been at the forefront to educate the public about IL. Although IL is important to society as a whole, it is more critical to students. University students, especially, routinely search and synthesize information from multiple sources for various knowledge-intensive tasks such as assignment presentations and report writing. Hence their academic performances are directly influenced by their IL skills (Loertscher and Woolls 2002).

However, today's youth are increasingly unwilling to initiate interaction with librarians when they encounter difficulties in information searching. They prefer to use information located from search engines despite the uncertain quality and reliability of such sources (Thomas, Crow, and Franklin 2011). On the one hand, they find traditional IL instruction in classrooms not engaging enough and expect to be entertained while being taught (Prensky

2003). On the other hand, digital games have become very popular, as games are able to provoke a deep sense of engagement (Atkinson 2002). Given the two trends among the youth, digital games present an opportunity for librarians to rethink the way to deliver IL education.

One way to utilize digital games in education is digital game-based learning (DGBL), which is learning by playing digital games (Prensky 2003). Digital games can serve as an engaging way to entertain players, build their cognitive abilities, encourage problem solving and collaboration, and increase self-esteem (Tan et al. 2011). Digital games can be so immersive that players often find themselves actively seeking information to solve the obstacles in games (Hamari et al. 2016). In fact, the behaviors of good game players conform to the IL framework by the Association of College and Research Libraries (ACRL), in which players “determine the initial scope of the task required to meet their information needs,” “organize information in meaningful ways,” and “synthesize ideas gathered from multiple sources.” Thus, DGBL has been found to be an effective way of educating the young generation (Atkinson 2002).

However, apart from the growing body of literature on using DGBL to teach lower-order thinking skills, concepts, and procedures, there is limited literature on games teaching higher-order thinking skills such as how to apply, analyze, or evaluate knowledge (Charsky 2010). Academic information searching is a higher-order thinking skill that involves synthesizing complex levels of knowledge and constructing meaning. Moreover, many educational games have failed because those who will actually use what is built (i.e., potential users) are often left out in the design process. Game designers typically assume that learning content can be easily combined with entertaining elements, thus turning a deaf ear to inputs from potential users (Gunter, Kenny, and Vick 2008). Consequently, the field of DGBL has seen many calls to hear from potential users and understand their needs to improve both educational effectiveness and game-play experience.

Game design is a complex problem where the solution and problem mutually define each other, so every proposed solution changes the understanding of the problem. One salient characteristic of such complex problems is that there are no stopping rules. Therefore, except in the early design stage, there is a need for continuous user involvement. The best guarantee for success is to go through iterative evaluations informed by user feedback (Salen and Zimmerman 2004).

This study aims to fill these research gaps by designing and evaluating an IL educational game, *Library Escape*. A user-centered approach was adopted, where potential users were actively involved as designers and testers. A participatory design (PD) workshop was first conducted, yielding a low-fidelity prototype. The game was later evaluated by other potential users and revised, resulting in a high-fidelity prototype. Next, design issues and challenges derived from this study are discussed. We conclude by presenting future directions of this game and research opportunities in this area.

Related Work

Digital Games in IL Education

Many well-known pedagogical principles can be found in DGBL game play (Prensky 2003). For example, the motivational and learning potential of DGBL can create positive impact on players as learners (Gee 2003). By infusing the elements of challenge, fantasy, control, feedback, and social interaction, DGBL immerses players in a virtual educational environment (Hamari et al. 2016). Such immersion is not only a key point for motivating players in the game but fundamental in maintaining coherence between the learning content and the game itself.

Academic libraries in the United States were among the first to use DGBL in IL education. The University of North Carolina reported success of its online board game called the Information Literacy Game, which was popular in promoting IL among students (Rice 2008). Arizona State University designed *Quarantined: Axl Wise and the Information Outbreak*, a digital game that simulates the complex processes of selecting, using, and evaluating multiple sources of information within a library setting (Gallegos et al. 2006). Moreover, the IL game from the University of Michigan, *Defense of Hidgeon*, obtained positive feedback from students (Markey, Leeder, and Rieh 2014). The game teaches students how to seek information, starting with finding background information and topic overviews, and continuing on to using library catalogs, databases, citation indexes, and other research tools. In its evaluation, most students reported positive learning experiences. Furthermore, Lycoming College developed a Flash-based digital game called *Secret Agents in the Library* to teach IL skills for composition to first-year students (Broussard 2012). To evaluate its effectiveness, a review quiz was administered to three classes, together with class discussions. Results showed that students who interacted with the game not only did better in the quiz but expressed increased enthusiasm and motivation in learning IL with such an innovative approach.

After an overview of available IL games, Mary Snyder Broussard (2012) formulated six best design practices for game designers: keeping it simple, having a plan to use it in class or market to faculty, “gating” key concepts, making it fun, giving feedback, and play testing throughout the development process. For example, gating for key concepts is an in-game device that prevents players from progressing before they master a skill. Using gating for the important concepts in DGBL ensures that students do what the game asks them to do. Moreover, feedback should be provided immediately and continuously during game play.

Participatory Design

Pioneered in Scandinavia in the 1970s, PD originated from the beliefs that design problems are sufficiently complex that no single person has all the knowledge required to solve them and that the design process would benefit from involving potential users, those who would be affected most by the design (Derboven, Van Mechelen, and Slegers 2015). In addition, the wide

use of computers indicates the necessity that any computer system must appeal to a broad base of users with varying levels of skill and preference. Potential users are given a voice in PD to communicate their ideas and concerns with educators, designers, and developers.

The PD approach has been applied in various contexts, from architecture to game design, for teaching social skills. For example, Socialdrome, an educational game for enhancing children's social skills, involved a group of 10-year-old children in the design process. The children played and tested the game prototype and developed new design concepts through storyboarding (Tan et al. 2011). The authors concluded that the active engagement in PD can stretch their creativity and critical thinking and enhance their cognitive understanding of social knowledge. The PD approach is not unfamiliar to libraries. It has been employed in situations ranging from redesigning study spaces and renovating library floors to reprogramming library services and restructuring library organizations (Foster 2011). For example, at the University of Rochester, PD was used to improve the library course page (Lindahl 2014). A group of students were invited to interact with a poster-size version of the course page to select items they liked and disliked. The librarians made immediate changes after the PD workshops.

User Evaluation

In addition to the active involvement of potential users in the early design stage, as the PD approach advocates, there is also the need to continuously involve users in the evaluation and development stages. There is, hence, an expanding body of literature on evaluation methodologies. Among them, user evaluation is considered the benchmark, because without involving users, the actual usage behaviors or usage scenarios cannot be accurately predicted (Ak 2012). For instance, in the design of Ergoshow, an educational software to teach ergonomics and health issues to primary school students, users' input helped designers adjust the difficulty level of questions, identify graphic and navigational design problems, and improve the wording and language (Rebelo and Filgueiras 2012). The authors concluded that enormous benefits to the project were achieved by bridging the communication gap between designers and potential users.

Based on the development stage, there are two main evaluation methods: formative and summative evaluation. Formative evaluation is carried out in the early design process, typically to address interface design, game play, and instructional issues with the intent to improve (Tan et al. 2011). Summative evaluation is conducted at the end, to examine the value and merits of the design. Because formative evaluation is carried out at an early stage, it can enhance the design substantially and prevent loss before a major error occurs. Further, representative users can help designers understand their expectations and focus on more important aspects of the design early on (Rebelo and Filgueiras 2012).

Method

Not using conceptual frameworks in educational digital games has significantly impeded games' adoption in formal learning contexts (Walsh et al. 2013). Therefore, we used the Information Search Process (ISP) Model (Kuhlthau 2004) as the theoretical framework to structure our game, because this model has been received favorably in organizing IL tutorials (Thomas et al. 2011). Although called an ISP Model, its scope is equivalent to most IL models and standards. Therefore, it is not merely an information-seeking process model but an encompassing IL education model (Thomas et al. 2011). The ISP Model divides a typical information-seeking process into six stages: task initiation, topic selection, prefocus exploration, focus formulation, information collection, and search closure. Negative feelings are common at the beginning, when the search problems are ill defined and the information seekers have little knowledge of what information is available. As the search progresses, the understanding of the search problem increases accordingly, resulting in a corresponding improvement in the information seekers' level of confidence. At the end of the search process, the information seekers will feel some relief and satisfaction when the required information is found, or they will feel disappointment and anxiety if it is not.

The method of the present study is comprised of two stages: the first is a PD workshop in the form of focus group discussion, which yielded a low-fidelity game prototype; the second stage is a user evaluation in the form of individual interviews, where the game prototype was further refined.

Participatory Design Workshop

Both undergraduate and graduate students were invited to represent the potential users in this study. A pilot study was carried out with three students (one undergraduate and two graduate students) to assess the feasibility of the protocol. After that, the formal PD workshop commenced.

To recruit participants, mass e-mails were sent out, and advertisements were posted at various locations in major local university campuses. To maximally explore different perspectives, students with diverse educational backgrounds in different stages of study were selected. Because the ideal size of a focus group is 8 to 10 participants (Tan et al. 2011), 7 university students were recruited for the PD workshop. They consisted of 3 men and 4 women between ages 21 and 27. There were 3 undergraduates (each majoring in sociology, marketing, and communication), 1 master's student (majoring in art history), and 3 doctoral students (each majoring in computer science, industrial design, and graphic design). All participants reported being experienced gamers, playing digital games at least 1 hour per week, with 2 playing more than 6 hours per week. As for their IL learning experience, all have searched for information and written assignments for academic purposes, and 1 participant attended a formal IL instruction by university librarians. The first author was involved as the

facilitator, whose main task was to introduce the study objectives, guide the discussion, and take notes. Other than being a game player, the first author was trained in interviewing techniques and had conducted interviews prior to this study.

The workshop was conducted in a focus group lab in a major university and lasted about 2 hours. All discussions were audio-recorded and carried out according to the instructions of the research protocol. At the beginning of the workshop, the participants were briefed on the research objective, which was to design a digital game to teach IL skills to university students. They were also instructed on the use of the ISP Model as the overarching structure and the necessary elements to develop an appropriate game, such as backstory, characters, rules, and goals.

To familiarize participants with the concept of DGBL, they were given 15 minutes to play two IL games, *Secret Agents in the Library* by Lycoming College and the *Information Literacy Game* by the University of North Carolina. The procedure is consistent with that used by Carl DiSalvo, Illah Nourbakhsh, David Holstius, Ayça Akin, and Marti Louw (2008). Next, their perceptions on what they liked and did not like about these games was discussed. Generally speaking, participants thought the two games could be improved. They suggested to vary the IL knowledge content, clarify game instructions, and improve the interface designs. They found the game graphics acceptable as they understood that librarians are not professional game designers. Next, they were introduced to the ISP Model and its stages. During the workshop, the participants were asked to jointly develop one game prototype as a team. This meant that they had to decide on genre, backstory, characters, setting, music, and so on and that consensus had to be reached for decisions involving these elements. Storyboarding was used, where blank paper and pencils were provided for participants to record their ideas. The workshop yielded a low-fidelity prototype.

User Evaluation Interviews

In the next step, individual user interviews were conducted to evaluate the low-fidelity prototype. The effectiveness of educational games is, on the one hand, judged by the games' capacity to improve learning performance and transfer the knowledge into other contexts (Atkinson 2002). It is also, on the other hand, judged by their provision of an enjoyable and engaging game-play experience (Gee 2003). Thus, the objective of our evaluation was to uncover perceptions of potential users on the game-play enjoyment and learning content of the low-fidelity prototype in order to further improve it.

To recruit participants, invitations were sent to participants who had attended the previous PD workshop and those who had not. While previous participants were given another opportunity to provide feedback on how the game had evolved, fresh perspectives could also be obtained from first-time participants. In total, 10 university students from a major local university were recruited; 4 had attended the previous PD workshop, and 6 were participating

for the first time. The group consisted of 6 men and 4 women between ages 21 and 29. There were 2 undergraduates (each majoring in journalism and banking), 4 master's students (each majoring in knowledge management, electronic engineering, computer engineering, and animation), and 4 doctoral students (2 majoring in computer science, 1 in information studies, and 1 in graphic design). Participants reported being experienced gamers, playing digital games at least 1 hour per week, with 3 playing more than 6 hours per week. All participants had searched for information and written assignments for academic purposes. The first author was involved as the interviewer, whose main task was to introduce the study objectives, present the game prototype, elicit feedback from participants, and take notes to capture participants' ideas accurately and in a timely manner.

All interviews were held in a meeting room and lasted about 1 hour. The 10 individual interviews took place over a 1-week period. At the beginning of each interview, participants were briefed on the research objective, which was to refine a low-fidelity IL game prototype. Afterward, the interviewer showed them the low-fidelity storyboards and explained all game scenarios and specific design decisions. The comments from participants were recorded and used to refine the game design and generate the next prototype.

Library Escape

This section first presents the low-fidelity prototype from the PD workshop and then the high-fidelity prototype based on the user evaluation interviews.

Low-Fidelity Prototype

Overview and Backstory

Because the game seeks to engage university students in learning IL skills, a role-playing adventure game was selected due to the genre's popularity among and attractiveness to youth. Participants believed that games in such a genre could increase the level of concentration and participation from players. After some discussion, they decided to name the game Library Escape.

There are three characters in Library Escape: Tom, the protagonist; Professor Babbage, a professor; and Ms. Senka, a librarian. Because there are six stages in the ISP Model, there are six missions in the game (see table 1). To add novelty and fun to an immersive game-play experience, each mission is set in a different venue (Guo and Goh 2014). Within each mission, Tom must complete the tasks assigned in the quests with the help of Ms. Senka. The difficulty of missions was arranged in an incremental manner to allow the development of knowledge from lower to higher level (Salen and Zimmerman 2004). To increase the sense of control by players in the game play, participants suggested that players can post their progress (leaderboard and status bar) and final e-certificates on some social media platforms (Harteveld and Sutherland 2015). Allowing the players to trumpet their success and participation to friends

Table 1. Venues and Missions

Missions: ISP Stages Venues	Topic	Quests (Objects Hidden In)
1: Closed stacks Task initiation	General introduction	1.1. Definitions of IL (two books) 1.2. The importance of IL to students (five fluorescent lamps) 1.3. ACRL IL standards (ruler) 1.4. Three modes of IL education delivery (whiteboard, television, and game console)
2: Reference desk Topic selection	Affect in information behavior	2.1. Basis for topic selection (telephone) 2.2. Library anxiety (clock pendulum) 2.3. The concept of uncertainty (notebook) 2.4. Possible factors contributing to library anxiety (mannequin, heart, sofa, skull, stool)
3: Open shelves Prefocus exploration	Information-seeking behavior	3.1. What is information seeking (crutch) 3.2. Wilson's First Model (Wilson 1981) (book) 3.3. Johnson Model (Johnson and Meischke 1993) (poster) 3.4. Ellis Model (Ellis 1993) (magazines)
4: Digital library Focus formulation	Kuhlthau's ISP Model	4.1. What is unique about the ISP Model (Kuhlthau's photo) 4.2. Different stages of ISP (staircase) 4.3. Focus on students' feelings (mirror) 4.4. Constructivist perspective (backward chaining) (map)
5: Café Information collection	Use of the ISP Model	5.1. Other works based on the ISP model (forward chaining) (ice machine) 5.2. How the ISP model helps students (flower) 5.3. How the ISP model helps librarians (gardener painting) 5.4. How the ISP model is used in libraries (coffee machine)
6: Exhibition area Search closure	Integrative session	6.1. When to stop looking for information? (stop sign) 6.2. Can information be "objective"? (sculpture) 6.3. Do people really "need" information? (painting) 6.4. "There is much more about IL than what is presented here!" (Ms. Senka)

and followers by posting the leaderboard and status bar on social media makes an attractive social reward for completing the game. One participant further suggested linking the social rewards with challenge level, "The game challenges could become easier if the players post the progress on Facebook, like Candy Crush."

The overriding goals of the game should be presented early and in a clear manner to grab players' attention (Salen and Zimmerman 2004). Therefore, the game begins with a short video clip that introduces the backstory. The grades of last semester have just been released,

and Tom, an excellent student, only managed a C for his IL module. Disappointed with the poor grade, Tom consults with Professor Babbage to find out the reason. Professor Babbage tells him that it is because his report has problems such as a lack of citations and the use of low-quality information sources. Being a motivated student, Tom asks how his grade can be improved. The professor ponders for a while, leads Tom to the basement of a deserted library building, and shoves him in. Professor Babbage says Tom could get appropriate IL training here, and she will be back in an hour to check on his progress (see top panel of figure 1). After Tom wanders in the library, an old woman suddenly appears. The woman is the senior librarian, Ms. Senka (see lower panel of figure 1), who turns out to be helpful in subsequent missions.

Mission 1: Closed Stacks

Mission 1 takes place in the library's closed stacks. The quests in Mission 1 teach foundational declarative knowledge ("what"), since such knowledge is usually easy to acquire and should be



Figure 1. Protagonist (upper panel) and librarian (lower panel)

taught prior to procedural knowledge (“how”) and causal knowledge (“why”) (Gagne et al. 2005). Tom has to obtain declarative knowledge such as the definitions of IL, its importance, and some international IL standards. To do so, Tom needs to explore his surroundings by interacting with objects to uncover questions and clues. All knowledge is hidden in these objects, which are only revealed when selected.

Tom walks up to a whiteboard and sees that the words on it read: “Tom, what does IL refer to? What skill set does an information literate person have?” Tom is taken aback because he does not know why the questions are addressed to him. Suddenly, a voice speaks up, “Don’t worry, the answer is within this room, you can explore the room to find the clues.” In *Quest 1*, Tom selects different objects to reveal the definitions of IL. The importance of IL skills (“why”) is explained in *Quest 2* so that Tom understands the point of going through the game. *Quest 3* drills deeper by outlining the ACRL’s IL standards, the widely adopted and de facto standards. Tom must interact with five hidden objects, each with one IL standard inside. In *Quest 4*, Tom needs to find the final object to know the three modalities to deliver IL education: face-to-face instruction, online tutorials, and DGBL. After that, Tom successfully completes this mission.

Mission 2: Reference Desk

Tom comes up to the second floor, where the reference desk is located. There are several objects: a telephone, a yellowish notebook, a broken clock pendulum, and an old computer. Tom is presented with an overwhelmingly long list of IL topics, along with ambiguous instructions. Not sure what to do, Tom initiates a conversation with Ms. Senka. After understanding Tom’s situation and his IL module grade, Ms. Senka recommends information-seeking behavior as the topic for his report.

In knowledge acquisition, learners need to understand basic concepts before solving actual problems (Kapp 2012); hence, Mission 2 introduces more relevant concepts. *Quest 1* explores the basis for topic selection after receiving an assignment. In *Quest 2* and *Quest 3*, Tom is introduced to the concept of library anxiety and uncertainty, two common feelings toward libraries. Afterward, Tom finds the hidden object for *Quest 4*, which further explains the factors contributing to library anxiety. Before leaving, Ms. Senka whispers to Tom, “Check out Z700 plus.” Tom wonders what the code is about. He enters the lift, and it takes him to the basement.

Mission 3: Open Shelves

Mission 3 takes Tom to the library’s open shelves. Mission 3 investigates the information in the shelves with the intent of finding a focus for the topic selected in Mission 2. This means that he has to do some initial investigation of this topic. In addition to shelves of books and magazines, there are some old posters on the walls. Tom browses the bookshelves aimlessly.

Suddenly he realizes that the code given to him in Mission 2 must refer to the code on the books. He rushes to the shelves with the Z700+ code and happily finds that Z700+ is where IL collections are located.

Now Tom needs to pick some useful titles from the collection. He is prompted to find out what information seeking is in Quest 1. Providing both relevant and irrelevant examples of concepts will enable the learners to clarify the differences between them and lead to in-depth understanding (Gagne et al. 2005). So in the subsequent three quests, Tom finds hidden objects that introduce three well-established but irrelevant models on this topic: Tom D. Wilson's First Model (Wilson 1981), J. David Johnson's model (Johnson and Meischke 1993), and David Ellis's Model (Ellis 1993). Tom needs to analyze and compare these models.

Tom is intimidated by the volume of information presented to him and does not feel confident that he can write anything on these models. Luckily Ms. Senka comes to rescue. She tells Tom not to worry, because such negative feelings are common when searching for information. The mission concludes, and a door is open for him to proceed to the next mission.

Mission 4: Digital Library

Tom enters a digital world. Mission 4 requires Tom to formulate a focus from the information encountered in previous missions. Tom realizes that the library has subscribed to thousands of e-books and e-journals. Because of the sheer volume of the material, he is overcome by fear. Ms. Senka cheers him up and teaches him some typical techniques.

Besides introducing relevant and irrelevant examples to illustrate a concept, providing a "best example" is also often used in concept teaching (Kapp 2012). The ISP Model can be considered the best example here, so it is the main focus in this mission. In Quest 1, Tom ponders the uniqueness of the ISP Model compared with others and why it is the best example in this context. Quest 2 and Quest 3 teach Tom the different stages described in the ISP Model. Quest 4 guides Tom to a deeper perspective of the ISP model by describing its constructivist background. Tom likes the ideas in the ISP Model; he thinks it is unique and wants to know more about it. Tom needs to explore the e-journals, e-books, and databases in the library website to gather such information. With the help and encouragement of Ms. Senka, Tom uncovers appropriate articles by using multiple searching techniques.

Mission 5: Café

Tom looks around and notices a café in one corner. There is a rusty coffee machine, ice machine, a broken piano, and some withered plants. Tom needs to collect more information on the ISP Model in Mission 5. He starts to gather information that extends and supports the model, which is hidden in the various objects.

The main focus in Mission 5 is to synthesize the information acquired from previous missions and understand how the ISP Model has been applied and received by professionals and students. Therefore, in Quest 1, Tom needs to consolidate the previously gathered information and understand other works that are based on the ISP Model. The hidden object in Quest 2 reveals how the ISP Model helps students during the information-seeking process. Tom relates it to his own experience in the game and how Ms. Senka has been helping him along the way. Similarly, Tom finds out how the ISP Model helps librarians to best provide assistance to students in Quest 3. In Quest 4, Tom learns the contribution of the ISP Model in libraries. Now Tom feels more confident about being able to complete the report on time.

Mission 6: Exhibition Area

After getting to the exhibition area on Level 2, Tom finds a big screen that is replaying the previous missions that he has completed. The first three quests in this mission ask Tom to think about “When to stop looking for information?” “Can information be ‘objective’?” and “Do people really ‘need’ information?” In the last quest, Ms. Senka tells Tom that “There is much more about IL than what is presented here” as an exhortation for Tom to explore IL further. The characteristic of the quests in Mission 6 is that there is no single right answer. Open-ended questions are situated on the highest level of question asking; thus, this mission requires Tom to make judgments about the entire information-seeking process in the game (Bloom 1956). Additionally, the open-ended questions can induce further discussion from the players after game play.

Afterward, Ms. Senka gives Tom two certificates showing his successful completion of all the tasks and his grade. The grade for the certificate is based on a final score aggregated from his performances in each quest. Professor Babbage also comes in to check on Tom and is very happy to see Tom's progress.

High-Fidelity Prototype

After the user evaluation was completed, participants' ideas were synthesized to further revise the low-fidelity design. To contrast how this version evolved from the previous one, we will organize the changes thematically rather than by mission, as in the description of the low-fidelity prototype.

Overview

The backstory remains largely the same except that the presentation format is changed from a video clip to comic strips (see figure 2). Participants from the user evaluation suggested that, although a video clip may afford a more immersive experience, it is also more likely to be skipped in order to start the game immediately. Moreover, comic strips have the advantage of



Figure 2. Backstory (revised)

being easier to design, clearer in presentation, and faster in viewing so that players quickly get an idea of the story. Another change is the gender of the librarian from a woman (Ms. Senka) to a man (Mr. Senka). This forces the players to confront gender stereotypes regarding librarians (Baylor and Kim 2009).

The ending of the game is also made more interesting. In the previous design, the game ended when Ms. Senka issues the certificates to Tom and Professor Babbage returns to congratulate him, which was “boring” according to participants in the evaluation. In the new version, Tom helps Mr. Senka restore the library to its former glory by finding a magic wand that used to belong to Mr. Senka.

The final reward is also changed. Participants deemed using two certificates, one graduation certificate and one grade certificate, as “unnecessarily complicated.” The revised design issues only one certificate. Furthermore, the grade is calculated based on the number of incorrect attempts (the fewer wrong attempts, the higher the grade). This can be shared on social media platforms.

Characters and Their Relationships

There are changes to how Tom meets Mr. Senka (see figure 3 for revised character design). After Tom is shoved into the dark library, he looks around and sees a dim glow. Tom summons his nerves and realizes that it is a ghost! He is about to flee when the ghost starts talking. His name is Mr. Senka, the guardian of the library. Unfortunately, he has lost his magic wand, and



Figure 3. Protagonist and librarian (revised)

his vision has failed so much that he cannot find it by himself. The magic wand is probably hidden somewhere in the library, and Tom can help by going around and finding it. In return, Mr. Senka can help Tom improve his IL grade. Meanwhile, Mr. Senka will accompany Tom along the way. At this, the two embark on a journey to find the long-lost magic wand.

The relationship between Tom and Mr. Senka becomes more intertwined. In the previous design, their relationship was simple: Ms. Senka gives guidance and encouragement to Tom when needed. This was criticized by some participants as “too simplistic,” making the librarian “insignificant and dispensable.” In the revised design, Mr. Senka takes on an additional role as the guardian ghost trapped in the haunted library, and Tom understands that he shoulders the responsibility of not only escaping the haunted library by himself but liberating the ghost. Mr. Senka can interpret the meanings of some objects that Tom does not understand. Moreover, to progress in the game, Tom will be presented with a map of the library at the end of each mission. He needs to select the next venue to investigate, given hints from Mr. Senka.

Missions and Quests

The six-mission structure is maintained in the revised game design. Because the ISP Model is used as the structure of the game, the missions rather than the venues are renamed after its six stages as a means of reinforcement. Moreover, the venues where each mission takes place are modified. Specifically, the exhibition area is replaced by an audiovisual collection to better reflect the collections and services provided by libraries. The sequence of venues is slightly adjusted to place similar types of collections together, from what students are most familiar with (i.e., physical books in closed and open shelves), to increasingly important resources (i.e., e-resources in digital libraries), to additional services offered by libraries (i.e., café).

There are changes in the learning content in some missions (see table 2). During the PD workshop, students were more engrossed with formulating game design ideas and less with the learning content. This is understandable as they were neither subject matter experts in the area of IL nor teachers who were well grounded in the pedagogical principles. Two remedies alleviated this problem. First, after the PD workshop, an expert on IL was consulted on the learning content. Second, in the user evaluation, the interviewer deliberately probed more on the pedagogical aspects of the game to gather feedback on how to incorporate the learning content in the game. As well, a deeper coverage of the subject matter replaces the wide breadth of coverage, and the content in each quest is bolstered. The result is that the number of quests in each mission is reduced from four to three to enable players to acquire more in-depth knowledge. Figures 4 to 9 illustrate the changes in graphic design of each mission.

Table 2. Venues and Missions (Revised)

Missions: ISP Stages Venues	Topic	Quests (Objects Hidden In)	Newly Added Objects
1: Task initiation Closed stacks	Starting out on your journey	1.1. Definition of IL (dictionary) 1.2. Importance of IL (wheel) 1.3. ACRL IL standards (scale)	Globe, hat, and coin
2: Topic selection Open shelves	Selecting your topic	2.1. Selecting the topic to work on (key) 2.2. Types of scientific literature (stool) 2.3. Understanding what library anxiety is and its causes (magazine)	Figurehead, balloon, and map
3: Prefocus exploration Reference section	Discovering what's in a library	3.1. Scholarly publication cycle (grandfather clock) 3.2. Searching Google Scholar (painting) 3.3. Using the OPAC (date stamp)	Telephone, trumpet, and cookies
4: Focus formulation Digital library	Looking for a needle in a haystack	4.1. Building blocks (scooter) 4.2. Pearl growing (search bar) 4.3. Successive fractions (torch light)	Mouse, binocular, and portion
5: Information collection AV collection	Focusing on what's important	5.1. Evaluating the articles retrieved (disk) 5.2. Information seeking as a form of information behavior (umbrella) 5.3. Reflecting on the ISP model (mirror)	Flag, poster, and owl
6: Search closure Café	Writing up	6.1. Roles and responsibilities of an author (flower) 6.2. Plagiarism (blackboard) 6.3. Citations (menu)	Guitar, crown, and magic wand



Figure 4. Mission 1 (revised)

Mission 1: Task Initiation. The first three quests from the previous design are retained, and the last quest on the different methods to deliver IL education is removed, because many participants remarked, “the answer to Quest 4 is too obvious.”

Mission 2: Topic Selection. As mentioned, the venue of Mission 2 is changed from the reference desk to the open shelves, because it is where physical books are displayed, following the closed stacks of Mission 1. The quests in Mission 2 see an increased difficulty level (Sweetser and Wyeth 2005). As a result, the topics of library anxiety and its causes are grouped. There is now an additional quest on the different types of scientific literature to help students understand the range of sources they can refer to when working on academic projects.

Mission 3: Prefocus Exploration. The previous quests introduced three irrelevant information behavior models to students. During the user evaluation, participants remarked that these models were “a digressing” because they were peripheral to IL. After some discussion of what the users wanted to learn, the revised quests are changed to: scholarly publication cycle, how to use Google Scholar, and the library online public access catalog. Participants remarked that the revised quests can teach them “more practical knowledge.”

Mission 4: Focus Formulation. Participants agreed that digital resources are increasingly important, but they felt the quests in the previous design “did not reflect anything related to digital libraries” and “they are similar to the quests in Mission 2.” Based on these comments and the knowledge that practical knowledge is preferred by students, we decided to teach practical information search strategies such as building blocks, pearl growing, and successive fractions in this mission.



Figure 5. Mission 2 (revised)

Mission 5: Information Collection. Elements of critical thinking are infused in Mission 5, and the game conforms more closely to the requirements of the ACRL IL standards, where information sources must be critically evaluated. It requires the players to evaluate the information retrieved and reflect on how the ISP Model has improved their information-seeking skills. Reflection is a critical part of the learning process in DGBL, as it affords a cyclic learning process such as sense making, reflecting, reaching conclusion, formulating strategies, and acting (Walsh et al. 2013).

Mission 6: Start Writing. Mission 6 mirrors the feeling of relief as described by the ISP Model, and it is situated in the library café, a place where library users can relax and chat. There is some modification in the quests in Mission 6. The previous quests were mainly open-ended and subjective questions intended to intrigue the players into in-depth thinking on



Figure 6. Mission 3 (revised)



Figure 7. Mission 4 (revised)

information seeking. However, in the user evaluation, participants expressed their concerns that such questions are “too philosophical and maybe even controversial.” This concern and the 1-hour time limit led to three new quests in this mission: the roles and responsibilities of an author, plagiarism, and the importance of citations.

Objects and Their Functions

In the previous design, a single object type was used to reveal all quests, which made the game linear and monotonous, and players who were more motivated to learn more or curious about IL were not able to explore further. So to encourage exploration of the learning content and to make the game more interesting, we adopted participants’ suggestion to add three new types of objects: explorer, bonus, and gateway objects. Explorer objects contain knowledge related to IL but are not relevant to answering the questions at the end of each mission. Bonus objects



Figure 8. Mission 5 (revised)



Figure 9. Mission 6 (revised)

contain no knowledge but can increase points, and gateway objects are used at the end of each mission to provide clues for the location of the next mission. The new objects can provide a wider range of stimuli in the game. We will describe how the added types of objects are used in Mission 1.

Explorer Object (globe). Tom finds out about how the concept of IL has evolved over the years to keep abreast of technological developments, principally the advent of the Internet.

Bonus Object (hat). If Tom walks over to collect it, he will get one additional point. The additional points can contribute to a better grade.

Gateway Object (coin). Tom needs to find it to proceed to the evaluation and next mission. He tries to read the engravings but cannot make any sense of them. Mr. Senka takes over the coin and helps him decipher the meaning.

Discussion

Game designers often make use of heuristics and design principles to navigate the complex design process. The two prototypes from this study yield many valuable insights on how the user-centered approach informs an IL game design project. Therefore, this section presents a discussion on three design insights obtained from the study.

Being Accommodating Yet Pragmatic

It is important to meet players' needs in game design (Sweetser and Wyeth 2005). As mentioned, many educational games fail because the designers turn a deaf ear to inputs from

potential users. The importance of the game staying relevant to players was emphasized by the participants in the PD workshop, one of whom noted, "In my opinion, if the game provides information that is relevant to me, I will be more interested to play it. But if you provide a topic in computer science, I will be confused."

However, this becomes difficult to achieve when different participants offer suggestions that contradict one another. For instance, in the PD workshop, two participants suggested making the game quests very difficult, while others opposed the idea. The two participants suggested borrowing ideas from Flappy Bird because "it is frustrating and challenging. When you play, you keep on advancing your skills. You will actually achieve more." Participants who were opposed to the idea argued that "that is a turnoff already to have it so difficult." Another participant recalled her own experience, saying that it "cannot be so challenging that one point it's over, two points it's over. . . . I played twice, no points at all. I was saying, 'what the hell? I will not play it ever again!'" Such contradictory insights from participants were resolved by asking them to elaborate on the pros and cons of their ideas, allowing the details of their ideas to emerge, and encouraging further discussion on which idea is better. In this case, the game could be relatively easy in the beginning, so that players will have confidence and interest to continue playing and learning. After that, the difficulty level will increase in each mission as the player progresses through the game and increases his or her skill level. By doing this, suggestions from both camps were accommodated.

Many scholars and practitioners have remarked that communication with potential users is challenging, because users lack technical knowledge, and system designs are typically expressed in highly technical terms (Kapp 2012). In our study, the participants did not have trouble articulating their game design ideas due to their educational background and the fact that they were all game players. The challenge they faced, however, arose out of their limited subject knowledge on IL education and on pedagogical principles.

This issue is especially highlighted in one user evaluation interview. More so than other participants, she held a pragmatic view of educational games and contended that players expect immediate and practical learning improvement and practical hands-on skills. Due to her training in journalism, where the most important points are always put in the first line of a story, she suggested that the most useful and important knowledge should be placed in the first mission to attract the attention of players, or "people will soon lose interest." Admittedly, her perspective is probably correct in marketing and media reporting; however, this potentially disrupts the flow of knowledge from easy to difficult. Many education principles, such as Benjamin Bloom's (1956) taxonomy, recommend that students understand the "what" (concepts) before the "how" (hands-on skills).

Given the imbalance in the knowledge base, the question regarding the extent to which input from participants should be accommodated and the extent to which researchers and

designers maintain their positions remains. We contend that PD is a model for involving users and understanding their needs in the design process, and the designer should be pragmatic in adopting suggestions and balancing different views, being aware of potential biases.

Balancing Enjoyment and Learning

Striking a balance between game enjoyment and learning is a challenging issue in educational game design (Van Eck 2010). Specifically, players will not be attracted to playing a game if there is too much learning and not enough fun, while the educational intent is lost if there is so much fun that players forget or fail to learn during the game play. In fact, this challenge was pointed out by one participant early in the PD workshop, "The learning content for us is very important. So no matter how fancy the backstory is, this learning content must be available in the game." Participants also understood the difficulty of achieving a good balance, stating, "You can make a game very interesting, but it becomes a big issue if you want to put learning inside." Nevertheless, the researcher constantly reminded the participants that any suggestions for making the game more enjoyable should not jeopardize learning, and vice versa.

To weave the learning content and game-play enjoyment, we used a captivating story. This allows the game rules to be incorporated into the story seamlessly so students can learn and acquire skills in the context the story provides (Guo and Goh 2014). The fictional story derived from the PD workshop was relatively simple: Tom is trapped in a library, and he can only escape by correctly solving all the puzzles in the six missions with the help from a librarian. During the user evaluation, participants pointed out that this story was "unconvincing" and "boring because players will just mechanically interact with the objects to uncover the IL knowledge." This pointed to the fact that the learning-enjoyment balance was tilted too much toward learning in the design.

Thus, in the user evaluation, while the learning content is retained, the element of enjoyment was increased by enriching the story. First, upon entering, a ghost asks the protagonist for help to find his magic wand in the library. In exchange, the ghost promises to provide help in learning IL knowledge along the way. The protagonist then has a dual mission in the game: to acquire IL knowledge for his own interest and to find the wand in exchange for the help in learning the IL knowledge. Second, the protagonist now needs to find clues that will activate the objects and to determine the sequence in which they have to be activated. In other words, players will no longer "mechanically" interact with the objects. The participants praised the revised game story to be, on one hand, "more gripping and suspenseful" and, on the other hand, "learning will be more fun."

From a practical point of view, we maintain that, although learning is the main objective of educational games, game-play enjoyment is critical because it makes learning more engaging and appealing. Stories and in-game activities can enable enjoyment and learning content to

coalesce. In designing Library Escape, some participants borrowed ideas from detective stories. So rather than just providing a few objects in which to hide the knowledge and forcing players to click on the screen mechanically, a series of clues are provided and players are encouraged to use logical reasoning to find appropriate answers. One participant gave an example of providing a “brush and mirror, so players need to collect the brush to dust the mirror to reveal something.” This participant also justified that this suggestion “will make the game more interesting, but will not distract too much on the learning, as it is not complicated.”

Being Guided yet Flexible

To be pedagogically effective, the content of educational games needs to be built on sound theoretical frameworks (Walsh et al. 2013). This enables practitioners to base their content on grounded principles rather than depend on intuition or individual experience (Beale and Creed 2009). However, scholars of game design have argued that players should be allowed to exercise control in the game, such as that over game characters and events (Sweetser and Wyeth 2005). According to this line of thought, game designers should hide the linear structure from players and allow them to freely probe and experiment in the game world. Players should also be given the freedom to choose difficulty levels so that they feel a greater sense of control. These opinions seem contradictory. On one hand, the pedagogy literature suggests that dictating the “correct” sequence is desirable, as missions that introduce new concepts can only be attempted after students have been exposed to necessary vocabulary and background knowledge. On the other hand, the literature of game design states that relinquishing control over to players is desirable.

Interestingly, participants voiced both viewpoints during the PD workshop. In Library Escape, the ISP Model was used to structure the game and provide guidance on how the librarian character could help players in the game. Some liked this structure because it “broke down the process of academic writing into smaller, more manageable steps. It was easy to understand”; they also felt that it made the game more “well organized.” The events in the game were prearranged to fit the six stages of the ISP Model. It is a linear game design where each mission neatly mapped to one stage in the ISP Model. Others suggested giving “more power to the users. For example, if I have completed two missions, I can ask other players to help me. It would make the game more engaging. Maybe I would play it longer.”

While using a theoretical framework can guide the flow of learning content in DGBL, it is not always possible, nor does it guarantee the game’s success. Designers should also give control and freedom to players in the game world to achieve a more immersive game play. In our work, we tried to reconcile the differences by fixing the overall structure of the game, in the form of six missions, but simultaneously allowing players to control the game within each mission. This means that players have to begin with the first mission and cannot skip

any missions or proceed to the next without completing the previous ones. However, in each mission, they can explore the venues and interact with various objects in any order. The level of acceptance of this idea was later validated with the participants in the user evaluation, and they were agreeable to it.

Conclusion

This article describes the design and evolution of an IL game, *Library Escape*, by actively involving potential users. Two user-centered approaches, PD, and user evaluation, were used to glean insights into users' likes and dislikes. In the PD workshop, participants created a low-fidelity prototype and constructed the basic game story. Thereafter, in the user evaluation, the low-fidelity prototype was assessed and refined. The feedback from the user evaluation was adopted to enhance and enrich the previous design, producing a high-fidelity prototype with improved game play and the learning experience. The findings indicated that involving potential users in the early design phase is a valuable and worthwhile practice. The study provides a good example of continuous dialogue with potential users.

The study makes several contributions. First, most IL games teach library-specific knowledge as such games are probably more customized for their students' needs. Hence, they have low generalizability and reusability because the content is closely tied with that library's practice and collections. Our study infuses a general theoretical model, the ISP Model, into an IL game design, thus making it reusable and generalizable by different libraries. Second, the game prototype and design ideas from participants can be used by other librarians as a starting point for exploration of DGBL in IL education. Moreover, the challenges and design insights from this study point toward the need for more critical thinking in user-centered design methods.

The study is not without limitations. First, the methods used in the PD workshop and user evaluation are focus group discussions and interviews. Other supplementary methods, such as user observations and game activity log analysis, should be used to triangulate the findings. Further evaluation with these methods will be carried out after the playable version is implemented. Second, this study only invited potential users (i.e., students) in the design process. They were game players but in entertainment-oriented games, and few had experience playing educational games. Hence, expert inspections with educators and game designer can be carried out as another source of feedback. Retrospectively, in the next round of game revision, the IL standards in Mission 1 will use a new IL framework, published shortly after the study was conducted (American Library Association 2015). It is our aim that after the game is completed, it will be made publicly available to interested parties such as librarians and instructors, to be used in classrooms as a supplemental IL teaching method. Nevertheless, the resulting two prototypes were received favorably, and findings from this study are sufficient to obtain preliminary understanding of the problems faced by educational game designers.

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